

Optimizing Hiring Criteria to Mitigate Employee Attrition Risk

Using a Logistic Regression Model

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MBAN625 - Enterprise Data Analytics

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September 26, 2026

Executive Summary

An upcoming project at IBM requires the Human Resource (HR) team to hire another batch of employees. To reduce cost and disruption in the project, the HR team aims to hire candidates with a low risk of attrition. Using the IBM HR Employee Attrition dataset that covers 1,470 employees, the team's analyst applied a Generalized Linear Model pathway to identify the key drivers of attrition probability. An intercept-only logistic model was compared with the logistic regression model using a leakage-safe, temporally ordered holdout set. The selected model achieved a precision of 0.50 and an accuracy of 0.84 at the operating threshold. However, the recall value of the model sits at 0.03, which may be due to other predictors showing no significant relationship with attrition. Evidence shows that the predictors that are statistically significant when relationships are revealed with employee attrition are distance from home (OR = 1.26, SE = 0.08, $p = 0.01$), number of companies worked (OR = 1.31, SE = 0.08, $p = 0.00$), and total working years (OR = 0.49, SE = 0.12, $p = 0.00$). Results also suggest that candidate employees from a Medical field have a lower attrition risk than those from the Human Resources field. It is recommended that the HR team incorporate these characteristics into the hiring criteria during the hiring process and use a two-proportion Z-test to assess the difference in employee attrition yearly. The HR team should use hiring criteria as a reference only and not as an absolute requirement when assessing the employability of each candidate.

Enterprise Problem and Objectives

Business Context, Stakeholder, and Decision

The Human Resources (HR) team at IBM is responsible for hiring new employees for an upcoming project. Because employee attrition is costly and disruptive, the team's analyst needs to decide which candidate characteristics to prioritize during the hiring process in order to reduce the risk of future attrition.

Problem Statement and Significance

Employee attrition leads to increased recruitment and training costs, loss of productivity, and reduced organizational knowledge. By identifying the key drivers of attrition probability, the HR team can make more informed hiring decisions and prioritize candidates who are less likely to leave within the first few years.

Objective and Analytical Question

The objective is to develop and validate a reproducible model that identifies key hiring factors associated with employee attrition risk. Analytical question: Which candidate attributes available at the time of hire accurately predict which employees will leave within the first few years?

Scope and Success Criterion

The analysis covers the 1,470 employees in the IBM HR Employee Attrition dataset. Each row represents one HR employee. Success requires at least one or more predictors in the model with $p\text{-value} < 0.05$.

Data Source and Preparation

Source, Permission, and Provenance

Data was obtained from Kaggle from the dataset IBM HR Employee Attrition Data, published by Lakshita Sharma. The dataset was identified as Version 1 on the Kaggle Data Explorer panel and was last updated in January 2026. It was accessed on September 19, 2026. Use was permitted under the CC0 Public Domain license.

Table 1

Dataset Profile

Element	Report Entry
Unit of analysis	One IBM HR employee per row
Raw and analytical sample	Raw sample: 1470 rows; Analytical sample: 1470 rows
Coverage	Population: 1,470 employees; Time period and geographic location: Not specified in documentation.
Target	Employee Attrition
Primary predictors/inputs	Distance from Home, Education Level, Education Field, Number of Companies Worked, Percent Salary Hike, Total Working Years, Training Times Last Year, Work Life Balance
Source and access date	Lakshita Sharma, Version 1, September 19, 2026, https://www.kaggle.com/datasets/lakshitasharma0404/ibm-hr-employee-attrition-data

Data Audit and Preparation

The dataset was inspected for missing values using `'pd.isna()'` to look for null values. To inspect the duplicates, `'pd.duplicated()'` was used. To look for values that may be invalid for analysis, the categorical values were inspected using `'pd.value_counts()'`. Finally, to check outliers,

boxplots were implemented. The columns that are shown to have outliers were further inspected by counting the values that are outside the upper and lower limits.

Table 2

Data-Quality Issues and Preparation Decisions

Issue checked	Evidence found	Action and rationale
Missing values	0 missing values	No further action is needed.
Duplicates	0 duplicates	No further action is needed.
Invalid values	0 invalid values	No further action is needed.
Outliers	3 variables contain outliers	The values are retained as there is no evidence that suggests data-entry error, as these values are plausible observations.
Leakage	Several variables are found to be only applicable to existing employees, and not employee candidates	These variables are removed.

Table 2 shows that the raw dataset contained 1,470 rows. There are no duplicate, missing, and invalid values. The raw dataset contained 35 columns. The following variables are dropped:

- Demographic variables, such as Age.
- Redundant values, such as Daily Rate.
- Variables that are only applicable to employees who are already employed, such as Years With Current Manager, because the target population are employee candidates who are still in the recruitment pool. These variables carry a risk of leakage.
- Stock Option Level, Department, Job Role, Job Level, and Monthly Salary because they are not based on the recruits, but the employers or both.

After dropping the aforementioned variables, the dataset is left with 9 columns.

Privacy, Ethics, and Data Limitations

No direct identifiers of any of the IBM HR employees, such as name and address, are included in the dataset. Variables, such as Gender and Age, are excluded because IBM is headquartered in America (EBSCO, 2013) and the U.S. Equal Employment Opportunity Commission (EEOC) states that discrimination against an applicant on the basis of race, color, religion, sex (including pregnancy, transgender status, and sexual orientation), national origin, age (40 or older), disability, or genetic information is prohibited. Marital Status is also excluded for the same logic as applied above.

The data that will be used for analysis is only a small reflection of the total characteristics that can be identified from employee candidates. The results of the analysis should only be used as support in the hiring process and not be used as a sole criterion.

Exploratory Analysis

Table 3

Illustrative Descriptive Statistics for Selected Numerical Variables

Variable	n	M	SD	Median	Range
Distance From Home	1,470	9.19	8.10	7.00	1-29
Education Level	1,470	2.91	1.02	3.00	1-5
NumCompaniesWorked	1,470	2.69	2.50	2.00	0-9
PercentSalaryHike	1,470	15.21	3.66	14.00	11-25
TotalWorkingYears	1,470	11.30	7.78	10.00	0-40
TrainingTimesLastYear	1,470	2.80	1.30	3.00	0-6
WorkLifeBalance	1,470	2.76	0.71	3.00	1-4

Note. n = count; M = mean; SD = standard deviation.

Table 3 shows that the distance from home appears slightly right-skewed, while several other variables (e.g., number of companies worked) also show mild right-skew, as their means are somewhat greater than their median. Education level, however, is an ordinal variable like work life balance and cannot be ideally described as normally distributed.

The standard deviation of the total working years in the context of the variable's range suggest that its spread may be somewhat more moderate relative to its wider observed range (0-40) unlike with the other variables, such as the distance from home that has a standard deviation of 8.10 with 1-29 range.

Figure 1

Illustrative Distribution of Attrition among Employees

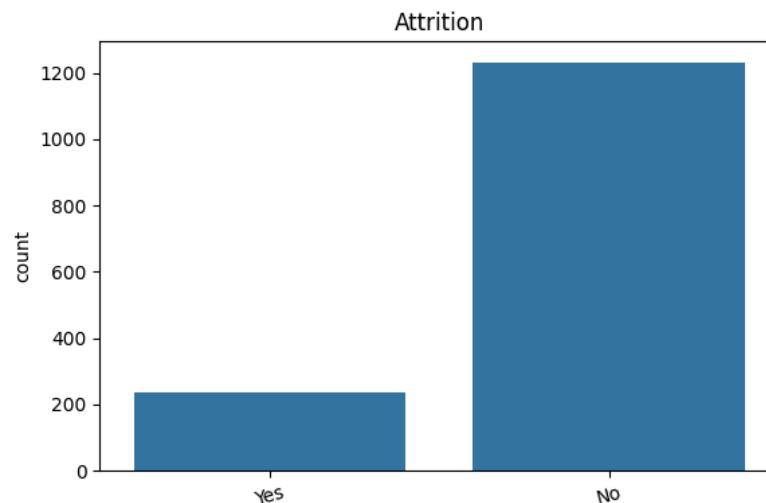


Figure 1 reveals that employees who have not attrited are about six times more than those who have.

Figure 2

Illustrative Distribution of Education Fields by Employees

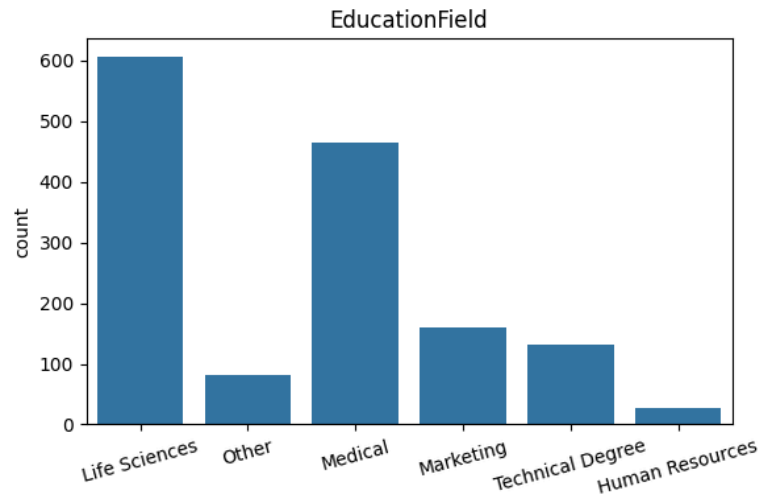


Figure 2 shows that employees who are in the “Life Sciences” field make up most of the sample, followed by employees who are in the “Medical” field. Employees from the “Human Resources” field are the least among the groups.

Analytical Pathway and Evaluation

Pathway B (Regression or Generalized Linear Model) was selected, specifically logistic regression, because the target variable (i.e., Attrition) is binary (Yes/No). Logistic regression is appropriate for modeling the relationship between a categorical binary dependent variable and multiple predictor variables (GeeksforGeeks, 2026). This approach supports identifying which candidate-related factors are most strongly associated with attrition and therefore may be most informative for hiring-related decision support.

Pathway B: Regression or Generalized Linear Model

The logistic regression model estimates the association between the predictors, which are distance from home, education level, education field, number of companies worked, salary hike

percentage, total working years, training times last year, and work-life balance, and the target, which is employee attrition. The baseline intercept-only logistic model was compared with the specified model. The data was then split into two sets: training (75%) and testing (25%) for validation performance examination.

Results and Discussion

Primary Findings

Table 4

Performance of the Logistic Regression Model against the Intercept-Only Logistic Model

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Intercept-only logistic model	0.84	0.00	0.00	0.00	0.50
Logistic regression model	0.84	0.50	0.03	0.06	0.69

Table 4 shows that both the baseline model (i.e., intercept-only logistic model) and the full model (i.e., logistic regression model) have the same accuracy, which means they make the same number of correct predictions overall. However, the advantage of the logistic regression model over the intercept-only logistic model is its ability to reduce the false positives as shown by its precision score of a 0.50 increase from 0.0. A 0.50 precision score means that the model will correctly identify approximately half of the employees who will truly either retire or resign. The glaring weakness of the model, however, is its Recall (0.03) and F1 score (0.06). It has no meaningful difference from the intercept-only logistic model. Recall value of 0.03 suggests that the logistic regression model has a 97% chance of falsely flagging an employee who will either retire or resign as not. Finally, the logistic regression model has a 0.19 increase in ROC-AUC

over the intercept-only baseline model, suggesting that it separates classes across thresholds better.

Table 5

Predictor Coefficient Results and Statistics Summary

Predictor	Coefficient	Standard Error	P> z	Odds Ratio
Distance From Home	0.23	0.08	0.01	1.26
Education Level	-0.10	0.09	0.26	0.91
Number of Companies Worked	0.28	0.08	0.00	1.31
Salary Hike Percentage	-0.02	0.09	0.79	0.98
Total Working Years	-0.73	0.12	0.00	0.49
Training Times Last Year	-0.09	0.09	0.30	0.91
Work Life Balance	-0.13	0.08	0.12	0.88
Education Field:				
Life Sciences	-0.49	0.25	0.05	0.93
Marketing	-0.12	0.17	0.47	1.15
Medical	-0.51	0.24	0.03	0.89
Technical Degree	-0.07	0.15	0.07	1.18
Other	-0.26	0.15	0.65	0.94

Focusing on which predictor is statistically significant ($p < 0.05$), Table 5 reveals that holding other variables constant, a one-standard-deviation increase in Distance From Home was associated with a 26% increase in the odds of employee attrition ($OR = 1.26$, $SE = 0.08$, $p = 0.01$), a one-standard-deviation increase in Number of Companies Worked was associated with a 31% increase in the odds of employee attrition ($OR = 1.31$, $SE = 0.08$, $p = 0.00$), and a

one-standard-deviation increase in Total Working Years was associated with a 51% decrease in the odds of employee attrition ($OR = 0.49$, $SE = 0.12$, $p = 0.00$).

The coefficients for the education fields represent differences in the log-odds of employee attrition relative to the Human Resources field, while holding other variables constant. This means that compared with Human Resources employees based on Table 5, Medical employees have estimated odds of attrition that are 11% lower, holding other predictors constant ($OR = 0.89$, $SE = 0.24$). However, it is important to note that the results for Medical employees is the only statistically significant result, having a p-value of 0.03, among the education fields.

Diagnostics, Errors, and Sensitivity

Diagnostics initially indicated perfect multicollinearity among the EducationField dummy variables due to redundant one-hot encoding. The Human Resources category was dropped to establish a reference baseline and eliminate the linear dependency. This allows the remaining field categories to be interpreted as relative odds ratios against the Human Resource field.

In terms of the Recall and F1 scores, the logistic regression model is not much different from the intercept-only logistic model with only a 0.03 and 0.06 increase respectively. These scores support the evidence of the weak relationships of some variables, such as Salary Hike Percentage, with Attrition.

Answer to the Analytical Question

Analytical results suggest that the pre-employment candidate characteristics that could be considered during the hiring process are the candidate's distance from home to work, number of companies worked, and total working years among the considered predictors. Total working years seemed to be the biggest factor among the studied predictors, as results suggest that if a candidate has one z-score above the average total working years, they are 51% less likely to resign or retire. Meanwhile, having more distance from home and the number of companies the candidate had worked before increases with the chance of the candidate's attrition. Furthermore, the results also suggest that a candidate from the Medical field is 11% less likely to resign or retire compared to a candidate from a Human Resources field.

Recommendations and Limitations

Prioritized Recommendation

Integrate the pre-employment candidate characteristics distance from home, number of companies worked, and total working years into the criteria of the hiring process. Furthermore, it is recommended to prioritize candidates from the Medical field over the Human Resource field if the candidate pool has a varied background. The Human Resource team of IBM should use these additional criteria for reference only and not as an absolute requirement.

Implementation, Monitoring, and Human Oversight

Table 6

Illustrative Implementation and Monitoring Plan

Action	Owner	Measure	Review rule
Hiring Criteria	Human Resource	Number of	Review quarterly; pause application and
Application	Team	Employees Hired & 90-Day Retention Rate	re-evaluate criteria if candidate pass-through or short-term retention falls below target thresholds.
Two-Proportion Z-Test	Data Analytics Team	Difference in Employee Attrition	Review annually; escalate to HR leadership for model recalibration if no statistically significant reduction in attrition is achieved or if subgroup disparities emerge.

Table 6 presents the implementation and monitoring plan for the new hiring criteria. First, the significant predictors should be integrated into the hiring process as part of the job interview or the candidate pooling. This should be reviewed quarterly to see the retention rate of the employees hired. If the results were unsatisfactory, the criteria should be re-evaluated. In the long term, the data analytics team should apply the two-proportion Z-test to see if there is difference between the employee attrition before the hiring criteria was updated and after. Afterwhich, it should be reviewed annually to see if it's subject to organizational change.

Limitations and Boundaries of Use

The model was validated on IBM's single snapshot of historical data and does not fully represent all characteristics of an employee. Furthermore, the population of this data was already employees of IBM, which is different from the target population the results intend to be applied

to, which are employee candidates during the hiring process. Another key structural limitation of this study is that the dataset represents employees exclusively within the Human Resources field. Therefore, the findings may not directly generalize to other enterprise departments (e.g., Medical, Marketing) that operate under different turnover baselines, compensation structures, and industry demand and to organization's changes. Interpretation, meanwhile, is limited to associations observed in the dataset and does not establish causal relationships. Future analysts should validate the model with an exhaustive list of employee candidate's characteristics available to the hiring team from varied hiring periods.

Reproducibility and AI-Use Disclosure

Notebook and Environment

The submitted notebook completed Restart Kernel and Run All on September 26, 2026, using Python 3.14.2 with kagglehub 1.0.2, matplotlib 3.11.1, numpy 2.5.1, pandas 3.0.5, seaborn 0.13.2, scikit-learn 1.9.0, and statsmodels 0.14.6. A fixed random seed of 29 was used where stochastic procedures were applied (dataset splitting). The notebook programmatically retrieves the permitted data and reproduces all principal tables, figures, and metrics reported here without personal paths, hidden files, or separate scripts.

Responsible AI Disclosure

The author used GPT-5.4 Mini to support conceptualization of the analytical pathway, planning, debugging, code explanation and editing. Representative assistance included comparison of different pathways, explanation of the analytical process such as model training, and appraisal of the primary findings. All suggested code was reviewed, executed, and compared with package documentation and notebook outputs. The team rejected the suggestion to exclude the factors Percent Salary Hike, Training Times Last Year, and Work Life Balance from the analytical

dataset because a hirer/employer can assess these characteristics from employee candidates through an interview. Final choices concerning data preparation, model specification, interpretation, citations, and recommendations were made and verified by the author. No confidential data or direct identifiers were submitted to an unauthorized AI service.

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